# 9.2 Two-Way Tables and Probability

Name _____ Period ____ Date _____

Algebra I 9.2 · Student sheet

**I can…**

- complete a two-way table.
- find joint and marginal relative frequencies.
- find conditional relative frequencies and use them to compare groups.

## Part 1 · Class survey

Copy the class results and fill in the totals.

| | Plays a sport | No sport | Total |
|---|---|---|---|
| Has a pet | | | |
| No pet | | | |
| Total | | | |

a. What percent of the whole class has a pet and plays a sport? _____

b. What percent of the students who play a sport have a pet? _____

c. Why are the answers to a and b different? _____

## Part 2 · Notes: fill in the blanks

**Joint frequencies** are _____ the table; **marginal frequencies** are the _____.

Joint relative frequency = cell ÷ _____ Marginal relative frequency = total ÷ _____

Conditional relative frequency = cell ÷ _____

The word “given” tells me which _____ to divide by.

## Part 3 · Examples (work them with the class)

### Example 1

A survey asked 80 students whether they own a pet. Complete the two-way table.

| | Owns a pet | No pet | Total |
|---|---|---|---|
| 9th grade | 24 | ? | 40 |
| 10th grade | ? | 12 | ? |
| Total | 52 | ? | 80 |

**Example 2**

Use the completed table.

a. What percent of the students are 10th graders who own a pet?

b. What percent of the students own a pet?

| | Owns a pet | No pet | Total |
|---|---|---|---|
| **9th grade** | 24 | 16 | 40 |
| **10th grade** | 28 | 12 | 40 |
| **Total** | **52** | **28** | **80** |

**Example 3**

Use the completed table.

a. Given that a student is in 9th grade, what is the conditional relative frequency that the student owns a pet?

b. Given that a student is in 10th grade?

c. Given that a student owns a pet, what is the conditional relative frequency that the student is in 9th grade?

| | Owns a pet | No pet | Total |
|---|---|---|---|
| **9th grade** | 24 | 16 | 40 |
| **10th grade** | 28 | 12 | 40 |
| **Total** | **52** | **28** | **80** |

## Part 4 · Partner practice

### Partner 1

Complete the two-way table about a new lunch menu.

| | Likes it | Doesn't | Total |
|---|---|---|---|
| **Male** | 18 | ? | 30 |
| **Female** | ? | 8 | ? |
| **Total** | **40** | **?** | **?** |

### Partner 2

Use your completed table from Partner 1. Given that a student is female, what is the conditional relative frequency that she likes the menu? Given male?